

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

01

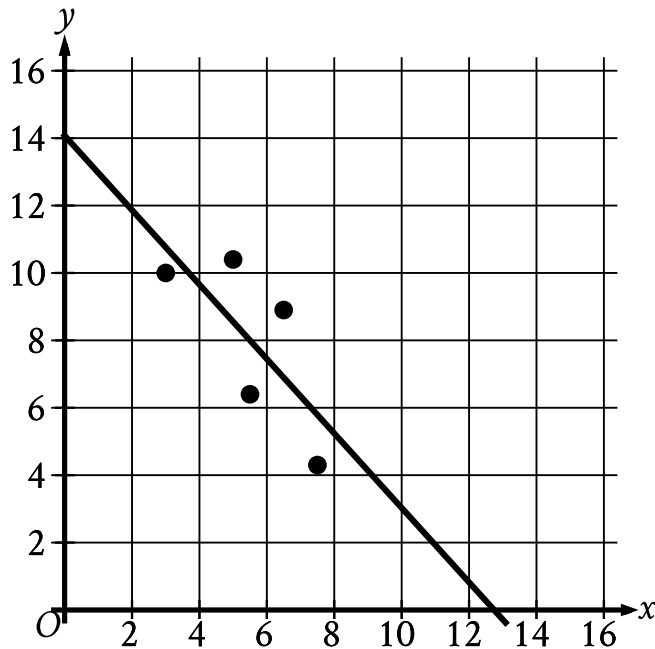
10 questions · ~15 min

Coat color	Eye color		
	Deep blue	Light brown	Total
Cream-tortoiseshell	16	16	32
Chocolate	12	4	16
Total	28	20	48

The data on the coat color and eye color for 48 Himalayan kittens available for adoption were collected and summarized in the table above. What fraction of the chocolate-colored kittens has deep blue eyes?

- A. $\frac{12}{48}$
- B. $\frac{12}{28}$
- C. $\frac{16}{32}$
- D. $\frac{12}{16}$

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
 - The line of best fit slants down from left to right.
 - The line of best fit passes through the following approximate coordinates:
 - (0 comma 14.1)
 - (12.7 comma 0)

Which of the following is closest to the slope of this line of best fit?

A. -3.3

B. -1.1

C. 1.1

D. 3.3

Q4

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

2, 10, 3, 7, 6

The mean of the list of numbers above is what fraction of the sum of the five numbers?

STUDENT-PRODUCED RESPONSE

☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

For the values j and k , the ratio of j to k is 11 to 12. If j is multiplied by 17, what is k multiplied by in order to maintain the same ratio?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A company that produces socks wants to estimate the percent of the socks produced in a typical week that are defective. A random sample of 310 socks produced in a certain week were inspected. Based on the sample, it is estimated that 12% of all socks produced by the company in this week are defective, with an associated margin of error of 3.62%. Based on the estimate and associated margin of error, which of the following is the most appropriate conclusion about all socks produced by the company during this week?

- A. 3.62% of the socks are defective.
- B. It is plausible that between 8.38% and 15.62% of the socks are defective.
- C. 12% of the socks are defective.
- D. It is plausible that more than 15.62% of the socks are defective.

Residents of a town were surveyed to determine whether they are satisfied with the concession stand at the local park. A random sample of 200 residents was selected. All 200 responded, and 87% said they are satisfied. Based on this information, which of the following statements must be true?

- I. Of all the town residents, 87% would say they are satisfied with the concession stand at the local park.
 - II. If another random sample of 200 residents were surveyed, 87% would say they are satisfied.
- A. Neither
 - B. I only
 - C. II only
 - D. I and II

Q8

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

An insurance company offers a series of three information sessions. 1,250 people attended the first information session. 72% of the people who attended the first information session attended the second information session, and 36% of the people who attended the first and second information sessions attended the third information session. How many people attended all three information sessions?

STUDENT-PRODUCED RESPONSE

--	--	--	--

.	/	.	/	.	.
---	---	---	---	---	---

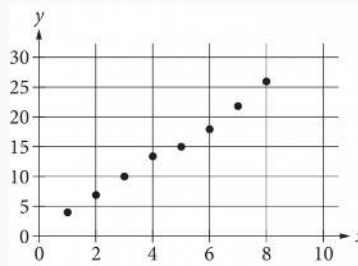
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Penguin Exhibit			
Type of penguin	Male	Female	Total
Chinstrap	41	59	100
Emperor	8	27	35
Gentoo	49	54	103
Macaroni	42	40	82
Total	140	180	320

The number of penguins in a zoo exhibit, sorted by gender and type of penguin, is shown in the table above. Which type of penguin has a female population that is the closest to being $\frac{1}{3}$ of the total female penguin population in the exhibit?

- A. Chinstrap
- B. Emperor
- C. Gentoo
- D. Macaroni



Which of the following could be the equation for a line of best fit for the data shown in the scatterplot above?

- A. $y = 3x + 0.8$
- B. $y = 0.8x + 3$
- C. $y = -0.8x + 3$
- D. $y = -3x + 0.8$